Free balancing equations worksheet worksheet 7

Maths | 9th grade | Algebra

1. If 3x + 5 = 2x + 9, what is the value of x?

A) 3
B) 4
C) 5
D) 6

2. What is the value of x if 4(x - 2) = 3x + 6?

A) 8
B) 10
C) 14
D) 18

3. Solve for x: 2x + 3(2x - 4) = 5x + 6.

A) 2
B) 3
C) 4
D) 5

4. Find the value of x if 7(x + 2) = 3x - 4.

A) -2
B) 0
C) 2
D) 4

5. What is the value of x in the equation 5 - 2(x + 3) = 2 - 4x?

A) -1
B) 0
C) 1
D) 2

6. Solve: 9(x - 1) = 3(3x - 5).

A) 3
B) 4
C) 5
D) 6

7. What is x if x/3 + x/2 = 5.

A) 6
B) 10
C) 12
D) 15

8. If the difference between twice a number and 3 is 9, what is the number?

A) 4
B) 6

C) 8
D) 12

9. What is the solution to the equation x/4 - 5 = 1?

A) 12
B) 20
C) 24
D) 28

10. Solve for x: 2(3x - 4) = x + 10.

A) 2
B) 3
C) 4
D) 5

Answer Key

1. A
Subtract 2x from both sides to get x + 5 = 9. Then, subtract 5 from both sides to find x = 4.

2. B
Distribute 4 to get 4x - 8 = 3x + 6. Subtract 3x from both sides to get x - 8 = 6, then add 8 to both sides to find x = 14.

3. D
Expand the equation to 2x + 6x - 12 = 5x + 6. Combine like terms to get 8x - 12 = 5x + 6, then subtract 5x from both sides to find 3x - 12 = 6. Add 12 to both sides and divide by 3 to find x = 5.

4. A
Distribute 7 to get 7x + 14 = 3x - 4. Subtract 3x from both sides to get 4x + 14 = -4, then subtract 14 and divide by 4 to find x = -2.

5. C
Distribute -2 to get 5 - 2x - 6 = 2 - 4x, simplifying to -2x - 1 = 2 - 4x. Add 4x to both sides to get 2x - 1 = 2, and then add 1 to find 2x = 3, so x = 1.5.

6. B
Distribute both sides to get 9x - 9 = 9x - 15. Subtract 9x from both sides to get -9 = -15, which is a contradiction. Therefore, there is no solution.

7. C
Find a common denominator and combine terms: 2x/6 + 3x/6 = 5, which simplifies to 5x/6 = 5. Multiply both sides by 6 to get 5x = 30, then divide by 5 to find x = 6.

8. B
Let the number be x. The equation is 2x - 3 = 9. Add 3 to both sides to get 2x = 12. Divide by 2 to find x = 6.

9. D
Add 5 to both sides to get x/4 = 6. Multiply both sides by 4 to find x = 24.

10. C
Distribute 2 to get 6x - 8 = x + 10. Subtract x from both sides to get 5x - 8 = 10. Add 8 to both sides to find 5x = 18, then divide by 5 to find x = 3.6.